

WoodWings — Developer Workflow & Debug Specification

1. System Overview

WoodWings is a Next.js 14 App Router project management system for a joinery/fit-out company. It manages projects from sales through fabrication to installation, tracks payments and warranties, and serves a multi-role team.

Stack

Layer	Technology
Framework	Next.js 14.2 (App Router, RSC)
Language	TypeScript 5.4
Styling	Tailwind CSS 3.4
Auth	jose JWT (HS256), 8-hour sessions
User store	SQLite via better-sqlite3
Business data	Airtable REST API
Client data	SWR 2.2 with auto-refresh
Charts	Recharts 3.8
Email	Resend (lazy-initialized)
Observability	In-memory metrics + Airtable log tables
Rate limiting	lru-cache (per-IP LRU)

2. Project Structure

```
woodwings/  
├─ app/  
│   ├─ api/  
│   ├─ dashboard/  
│   ├─ home/  
│   ├─ login/  
│   └─ admin/  
├─ lib/  
├─ components/  
├─ data/  
├─ scripts/  
└─ middleware.ts
```

3. Environment Setup

.env.local

```
AIRTABLE_API_KEY=patXXXXXXXXXXXXXXXXX  
AIRTABLE_BASE_ID=appXXXXXXXXXXXXXXXXX  
SESSION_SECRET=your_64_char_secret  
  
RESEND_API_KEY=re_XXXXXXXXXX  
ACCOUNTANT_EMAIL=accountant@woodwings.com  
  
NEXT_PUBLIC_BASE_URL=https://yourdomain.com
```

Install & Run

```
npm install  
npm run dev  
npm run build  
npm run start
```

4. Data Architecture

Dual Storage Model

Store	Content	Technology
SQLite	Users, roles, passwords	better-sqlite3
Airtable	Projects, tasks, payments, logs	REST API

Rule: No cross-usage between stores.

5. Auth System

Login Flow

```
POST /api/auth/login
→ Validate (Zod)
→ Rate limit check
→ SQLite user lookup
→ bcrypt compare
→ JWT creation (8h)
→ Set cookie
→ Redirect
```

Role Map

Role	Dashboard
superadmin	/dashboard/superadmin
manager	/dashboard/mgr
sed	/dashboard/sed
fabrication	/dashboard/fab
installation	/dashboard/fix

6. Development

Dev Loop

```
npm run dev
```

Type Check

```
npx tsc --noEmit
```

Production Build

```
npm run build
```

7. Key Patterns

API Route Pattern

```
export const POST = requireRole('manager')(async (req) => {  
  const body = await req.json()  
  const parsed = Schema.safeParse(body)  
  if (!parsed.success) return error  
  return success  
})
```

SWR Pattern

```
const { data } = useSWR(  
  condition ? '/api/data' : null,  
  fetcher,  
  { refreshInterval: 30000 }  
)
```

8. Common Problems

- **SQLITE_BUSY:** Kill duplicate node processes

- **Missing env vars:** Check .env.local
- **Session secret error:** Must be ≥ 32 chars
- **Airtable 403:** Use correct PAT + base access
- **Wrong dashboard redirect:** Fix role value
- **CSRF error:** Set correct base URL
- **Rate limit 429:** Reduce request bursts

9. Observability

Health Endpoint

GET /api/health

Logs

[INFO] request success
[API Error] failure
[MetricsSnapshot] issues

10. Security

Protection	Implementation
Password hashing	bcrypt (cost 12)
JWT	HS256, 8h TTL
Cookies	httpOnly, SameSite=lax
Rate limiting	5 attempts / 15 min
Validation	Zod
CSRF	Origin check